

Preliminary

This chapter covers the essential background needed for the study of linear algebra. Feel free to skip sections that you are already familiar with. This chapter serves only as a quick recap of the necessary mathematical background, so it might not be the best place to start if you are fresh to these topics. If you require a more detailed construction of these topics, I recommend checking out the first chapter of *Topology* by James Raymond Munkres (2nd ed.).

In addition to the following knowledge presented in this chapter, basic knowledge of logic and proof techniques is recommended, including:

- Direct proof
- Proof by contradiction
- Proof by induction
- Proof by contrapositive
- Proof by cases (exhaustion)
- Constructive proof (constructive existence)
- Non-constructive existence proof
- Uniqueness proof
- Proof of equivalence (iff proof)
- Disproof by counterexample
- Pigeonhole principle